

FIELDCELL · DESIGN DECISION RECORD

FieldCell recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	FCL-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/fieldcell		
Source	docs/decisions/0002-recommendations-accepted.md @ f0663ca		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all open recommendations and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for items A1 to A7; items O1 to O3 remain proposed

■ Context

After the TRL 3 session, docs/REVIEW.md (session 2026-09-25) listed ten items as "Proposed, awaiting Amish", seven of them with a recommendation, and FCL-DDR-001 left items O1 to O5 open. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every open item that carries a recommendation is therefore decided in favor of it; where the recommendation chose among options, the recommended option is the decision. Items without a recommendation stay open. TRL 4 remains on hold by Amish's instruction, so any part of a decision that needs building, testing or purchasing is recorded but not done.

■ Options considered

The options for each item are those listed in docs/REVIEW.md , session 2026-09-25, "Proposed, awaiting Amish", items 4 to 10, and in FCL-DDR-001, Table 2.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
A1	Budget (R12); REVIEW item 5	Option (a): raise <code>budget_usd</code> to about \$2,100 (BOM plus a small contingency), to be confirmed by quotes for the pack, PV wings and inverter	<code>project.yaml</code> <code>budget_usd</code> 1,500 to 2,100; R12 target \$2,100; BOM \$2,051 after A4 and A5, so R12 is now met with about 2.4 % contingency. Obtaining quotes is purchasing work and is on hold with TRL 4
A2	Mass limit (R6); REVIEW item 6	Option (a): relax the R6 mass limit from 70 kg to 75 kg, keeping the steel frame and flat-free tyres	FCL-REQ-001 R6 target 75 kg. The cart is 75.06 kg after A4 and A5, so R6 is still not met, by 0.06 kg; a follow-up is proposed in <code>docs/REVIEW.md</code>
A3	Electronics box ventilation (R8, R9); REVIEW item 7	Adopt the thermostat-controlled IP54 filter fan (40 to 60 m ³ /h) with IP54 exhaust filter	Already priced (BOM item 7) and modeled; now marked decided in FCL-PRC-001, FCL-REQ-001 (R8 wording) and the BOM
A4	Sun shade (R9); REVIEW item 8	Add a light reflective shade over the battery and electronics enclosures; obtain inverter derating data at 45 °C	New BOM item 18 (\$30, 1 kg); shade added to <code>cad/src/model.py</code> and the media; FCL-CAL-001 section G: pack in noon sun at 45 °C ambient 55 to 48 °C, box at 1 kW 54 to 53 °C. The derating data comes with supplier selection and is on hold with the quotes
A5	Hinge spacer; REVIEW item 9	Raise each hinge line by a 20 mm spacer	<code>hinge_z</code> 470 to 490 mm; BOM item 14 \$45 to \$60 with two 20 × 20 × 2 mm aluminium angles (0.53 kg); wing to tyre clearance 8 to 28 mm; stowed height 1,170 to 1,190 mm; drawing FCL-DWG-002 Rev P1 to P2
A6	"Peak sun hours" basis (R3, R4); REVIEW item 10	Confirm the reading as daily global horizontal irradiation in kWh/m ² /day	R3 and R4 targets restated in kWh/m ² /day GHI; the assumption no longer marked proposed
A7	Heavy parts low and over the axle; FCL-DDR-001 O4, REVIEW item 4	Keep it as a layout rule	FCL-PRC-001 key design choice marked decided

Table 2. Items that remain open (Proposed, awaiting Amish).

#	Item	Why it stays open
O1	Whether to seek a responder organization to review the deployment sequence and load profile, and which one	No recommendation was made
O2	How FieldCell relates to PowerBox and SwapCell in the portfolio	No recommendation was made
O3	Whether to add an AC charger or a vehicle 12 V input for cloudy periods	Listed as an open option with no recommendation

■ Consequences

- Controlled documents revised: FCL-PRB-001 v0.4, FCL-PRC-001 v0.4, FCL-REQ-001 v0.4, FCL-CAL-001 v0.2 and FCL-DDR-001 v0.2. The model, STEP and STL files, drawing FCL-DWG-002 (Rev P2), BOM and concept media were regenerated.

- Requirement status (FCL-CAL-001 v0.2): 7 met (R1, R2, R3, R4, R5, R10, R12), 1 not met (R6, by 0.06 kg), 2 at risk (R7, R9), 2 not verifiable at TRL 3 (R8, R11). Before: 6 met, 2 not met (R6, R12).
- The decisions add 1.5 kg. With the relaxed limit the cart sits at 75 kg, so R7 pull force on grass is 147 N against 150 N.
- No cross-repo change follows from these decisions.
- `trl` and `trl_target` stay at 3. TRL 4 work (quotes and purchasing, building, testing) stays on hold by Amish's instruction.